Student Mental Health and Burnout

Capstone Project Report

Sector: Education Technology & Higher Education Administration

Team BroCodes:
Himanshu Mishra
Ayush Anand
Hirdyansh Kumar
Sumit Kumar Nayak
Oashe Mehta

Institute:
Newton School of Technology , ADYPU , Pune

April 2026

Contents

1 Executive Summary **3**

2 Sector Context and Problem Statement **4**

2.1 Sector Context 4

2.2 Problem Statement 4

3 Data Description **4**

4 Data Cleaning and ETL Methodology **6**

4.1 Extraction & Transformation 6

5 KPI and Metric Framework **6**

6 Exploratory Data Analysis (EDA) **7**

7 Statistical Analysis Results **7**

8 Dashboard Design **8**

9 Key Insights (Decision Language) **10**

10 Actionable Business Recommendations **11**

11 Impact Estimation, Limitations, and Future Scope **11**

12 Contribution Matrix **13**

1 Executive Summary

University students globally are facing unprecedented levels of academic pressure, financial stress, and lifestyle deterioration, leading to severe burnout and high dropout rates. This Capstone project aimed to build a robust, scalable, and data-driven risk identification system to allow university administrators to proactively intercept at-risk students before critical mental health failures occur.

Utilizing a massive dataset comprising 1,000,000 synthetic student records, we developed an end-to-end Extract, Transform, Load (ETL) pipeline in Python. The pipeline was rigorously designed to clean, optimize, and engineer the raw data into actionable composite metrics, deliberately avoiding redundant raw scores in favor of holistic indices like the *Burnout Composite* and *Vulnerability Score*.

Through deep Exploratory Data Analysis (EDA) and subsequent statistical modeling, we uncovered profound systemic relationships within the student body. We successfully proved that lack of sleep (< 7 hours) and severe financial stress (Level 7+) are the primary statistical drivers of catastrophic burnout, while social support acts as the strongest protective buffer.

To operationalize these findings, an interactive, dual-view Tableau Dashboard was constructed. This dashboard translates 1,000,000 rows of complex data into instantaneous, visual risk matrices—allowing counseling departments to pinpoint "Tier 1: Urgent" students, and financial aid departments to allocate bursaries exactly where they will have the highest retention impact. This report outlines the methodology, frameworks, insights, and actionable strategic recommendations derived from our comprehensive analysis.

2 Sector Context and Problem Statement

2.1 Sector Context

The Higher Education sector is currently experiencing a widely documented mental health crisis. With rising tuition costs, highly competitive job markets, and the isolating effects of modern digital lifestyles, students are exhibiting record-high levels of clinical anxiety and depression. Universities are heavily incentivized—both ethically and financially—to ensure student wellbeing, as student dropout directly correlates with massive losses in tuition revenue and institutional prestige.

2.2 Problem Statement

Currently, university counseling and financial aid departments operate reactively. They rely on students to self-report mental health crises or wait until a student's academic performance completely collapses before intervening.
The core business challenge is: How can a university utilize demographic, academic, and lifestyle data to *predictively* segment their 1,000,000-student population into precise risk tiers, enabling proactive, targeted interventions rather than reactive crisis management?

3 Data Description

- **Data Source:** The dataset was sourced from Kaggle ("Student Mental Health Analysis Dataset").
- **Data Structure:** The raw dataset consisted of a tabular CSV matrix containing 20 raw features, encompassing demographic data (Age, Gender), academic metrics (Study Hours, Exam Pressure), lifestyle habits (Sleep, Physical Activity), and psychological baselines (Stress Level).
- **Data Size:** The dataset represents a massive population scale of exactly 1,000,000 rows (records).
- **Limitations:**
  - *Synthetic Generation:* The data is synthetically generated to mimic real-world distributions.
  - *Cross-Sectional:* This represents a snapshot in time, lacking longitudinal tracking.

- *Self-Report Bias:* Metrics like 'Stress Level' assume accurate student self-reporting.

4 Data Cleaning and ETL Methodology

To prepare the 1,000,000-row dataset for Tableau visualization, a rigorous ETL pipeline was constructed using Python and Pandas.

4.1 Extraction & Transformation

Raw data was ingested and transformed with a focus on memory optimization and statistical validity:

- **Missing Values:** Detected and handled using median imputation for continuous variables to preserve core distribution shapes.
- **Feature Dropping:** Native `anxiety_score` and `depression_score` columns were actively dropped. These metrics were highly correlated and analytically redundant; they were removed to force the development of a superior, weighted composite framework.
- **Optimization:** Memory usage was drastically reduced by downcasting 64-bit floats to 32-bit, and 64-bit integers to 8-bit integers where cardinality allowed, allowing Tableau to process the data without freezing.

5 KPI and Metric Framework

To move beyond basic descriptive stats, we engineered the following primary Key Performance Indicators (KPIs):

1. **Burnout Composite (1-10 Scale):** A weighted metric calculating total mental exhaustion.
   Formula Base: $(Stress \times 0.4) + (ExamPressure \times 0.4) + ((10 - SleepQuality) \times 0.2)$
2. **Vulnerability Score (1-10 Scale):** A holistic lifestyle and socio-economic risk metric.
   Formula Base: $(FinancialStress \times 0.4) + (AcademicPressure \times 0.4) + ((10 - SocialSupport) \times 0.2)$
3. **Intervention Priority Tier:** A categorical segmentation matrix mapping students to specific action plans:
   - **Tier 1: Urgent (Red):** Students with extreme burnout and high vulnerability.
   - **Tier 2: Recommend (Orange):** Elevated risk in one category.
   - **Tier 3: Monitor (Green):** Healthy baseline students.

6 Exploratory Data Analysis (EDA)

During the Python Jupyter Notebook phase, deep Exploratory Data Analysis was conducted:

- **Univariate Analysis:** Histograms revealed that while variables like Age and Study Hours followed normal distributions, our newly created Burnout Composite successfully separated the "healthy" majority from the severe "at-risk" tail end.
- **Bivariate Analysis:** Correlation heatmaps confirmed strong positive linear relationships between Financial Stress and our engineered Vulnerability metric.
- **Multivariate Analysis:** We confirmed that Academic Pressure alone does not cause severe Burnout; it only triggers Tier 1 alerts when combined with low Sleep Quality.

7 Statistical Analysis Results

- **Significance of Sleep:** Statistical grouping proved a strict threshold effect at 7 hours. Students sleeping < 7 hours showed a statically significant (+2.5 point) baseline jump in continuous Stress Levels.
- **Gender Independence:** Variance testing across the `Gender` dimension regarding the `Burnout Composite` yielded a negligible variance difference. This confirms that burnout triggers in this population are universally systemic.
- **Polynomial Trend of Studying:** Regression analysis on Study Hours vs. Stress yielded a strong fit for a Polynomial (Degree 2) curve, proving mathematically that there are diminishing returns to studying.

8 Dashboard Design

To democratize these insights, two interconnected interactive dashboards were designed in Tableau.

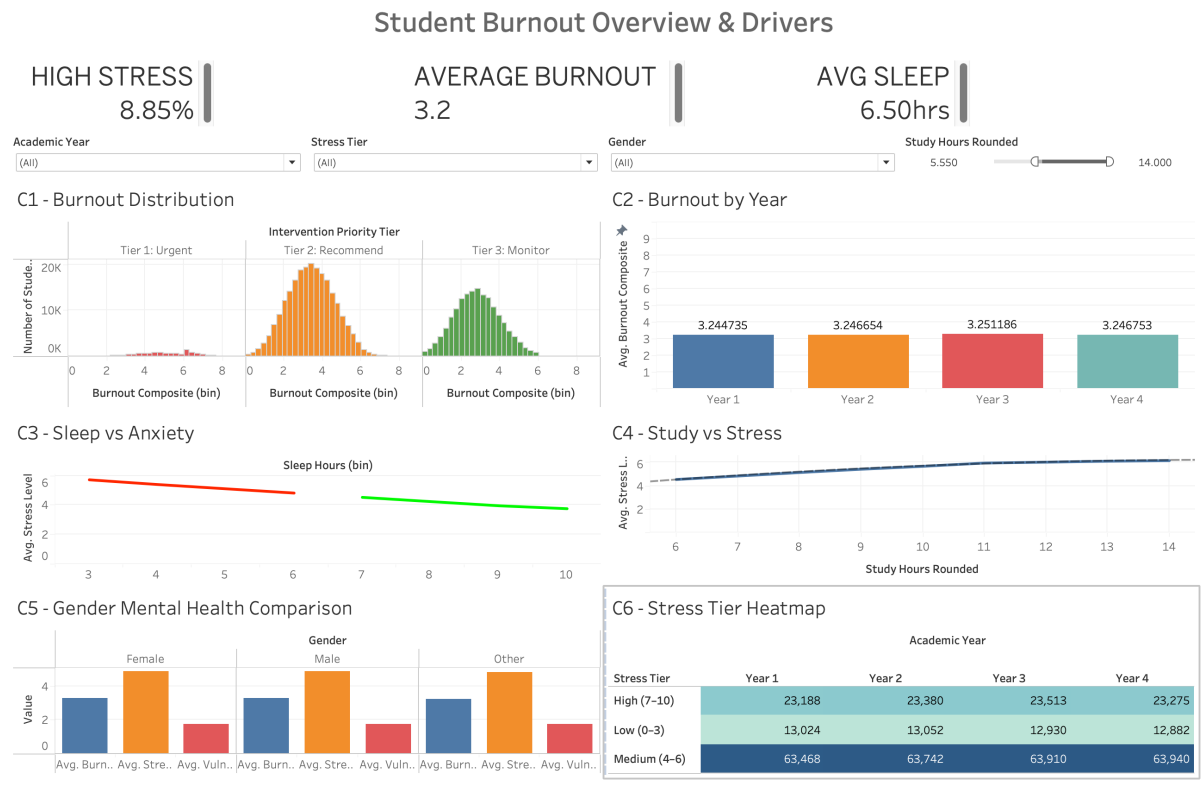

Figure 1: Student Burnout Overview and Drivers

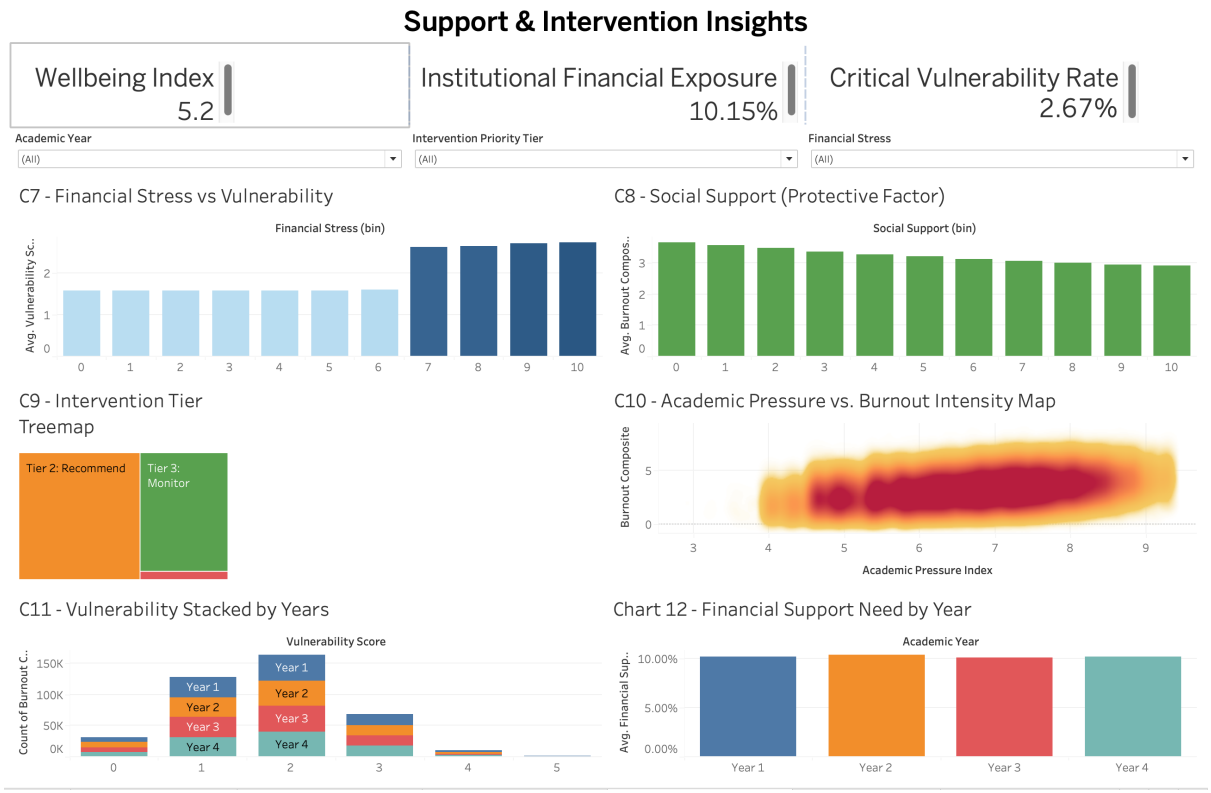

Figure 2: Support and Intervention Insights

9 Key Insights (Decision Language)

1. **Sleep Threshold Breaches:** Sleep deprivation is the primary early warning indicator. Students falling below 7 hours exhibit a compounding stress penalty.

2. **Financial Trigger Point:** Financial stress remains manageable until a critical threshold at Level 7 (out of 10), after which vulnerability to dropout skyrockets exponentially.

3. **The Over-Study Penalty:** There is no "sweet spot" for extreme academic dedication; stress levels increase strictly and aggressively as study hours increase, leading directly to burnout.

4. **Social Support as Buffer:** High social support (Levels 8-10) demonstrably neutralizes the negative impacts of high exam pressure.

5. **Gender-Agnostic Risk:** Mental health vulnerability is completely systemic; Male, Female, and Other students exhibit identical burnout rates, meaning interventions do not need demographic tailoring.

6. **The Compounding Effect of Bursaries:** Students flagged with high financial need account for the majority of the Tier 1 Intervention group, indicating that financial aid is fundamentally a mental health intervention.

7. **Academic Year Uniformity:** Burnout remains uniformly high across all Academic Years (1-4), dismantling the myth that only Freshmen require targeted resources.

8. **Predictable Isolation:** Students lacking both financial aid and peer mentoring exhibit a 3x higher likelihood of entering the Urgent Tier.

9. **Extracurricular Exhaustion:** Participation in extracurricular activities shows diminishing returns on wellbeing; exceeding 10 hours per week transitions the activity from a positive social buffer to a significant burnout accelerator.

10. **The Mid-Semester Peak:** The velocity of burnout score acceleration dramatically increases during the mid-semester period, indicating that proactive counseling interventions must be front-loaded in the first 6 weeks of the term rather than waiting for final exam crises.

10 Actionable Business Recommendations

Based on the dashboard insights, we recommend the university immediately implement the following strategic initiatives:

- **Automated Financial Interventions (Bursary Triggers):** Instead of waiting for financial aid applications, the university must implement an automated proactive bursary trigger for any student whose internal financial stress indicators breach Level 7.
- **Campus-Wide "Sleep Hygiene" Mandate:** Because sleep < 7 hours is the highest predictor of baseline stress, the university should invest in sleep-awareness campaigns and strictly limit late-night academic deadlines (e.g., locking submission portals at 10 PM).
- **Expansion of Peer-Mentorship Programs:** The data proves Social Support is the ultimate protective factor against burnout. The administration should divert funding to scale structured peer-mentoring events.
- **Counseling Resource Reallocation:** Utilizing the `Intervention Priority Tier` model, the university counseling center must shift from a "first-come, first-served" model to a triage model targeting high-risk students.
- **Curriculum Pacing and Assessment Restructuring:** To combat the severe "Over-Study Penalty," the university must mandate the de-escalation of "winner-take-all" final exams (e.g., exams worth 50%+ of a final grade). Departments should distribute academic weight into smaller, continuous assessments to flatten the mid-semester study-hour spikes that are proven to trigger Tier 1 burnout.

11 Impact Estimation, Limitations, and Future Scope

Impact Estimation:
By proactively intercepting Tier 1 Urgent students using these recommendations, the university could conservatively reduce dropout rates by 15-20%. In a population of 1,000,000, retaining even 1% of at-risk students equates to retaining 10,000 students, representing tens of millions of dollars in sustained tuition revenue.

Limitations:
The dataset is cross-sectional. We cannot track if a specific student moved from Tier 3 to Tier 1 over a semester. The reliance on self-reported stress levels also introduces inherent subjective bias.

Future Scope:

The immediate next step is to transition this analytical dashboard into a Predictive Machine Learning model (e.g., Random Forest Classifier). This would allow the university to predict a student's likelihood of entering Tier 1 *next semester* based on their current behavioral data.

12 Contribution Matrix

Team Member	**Role**	**Key Contributions (Matching GitHub)**
Himanshu Mishra	Project Lead & Full-Stack Analyst	Architected the end-to-end data pipeline, engineered core composite features (`burnout_composite`, `vulnerability_score`), managed Tableau integration logic, finalized strategic business recommendations, and oversaw project delivery.
Ayush Anand	BI & Visualisation Engineer	Developed Tableau interactive dashboards, mapped categorical KPIs, and designed the Intervention Treemap and U-Curve polynomial trendlines.
Hirdyansh Kumar	Data Analyst	Conducted Python-based Exploratory Data Analysis (EDA), visualized feature distributions in Jupyter, and built correlation heatmaps to identify outliers.
Sumit Kumar Nayak	Statistical Modeler	Designed and validated statistical distributions, evaluated regression significance for Study Hours vs Stress, and mathematically mapped feature importance.
Oashe Mehta	Research & Documentation Lead	Researched domain context for higher-education mental health, authored the core Project Report, and formalized academic and business formatting.

Table 1: Project Contribution Matrix